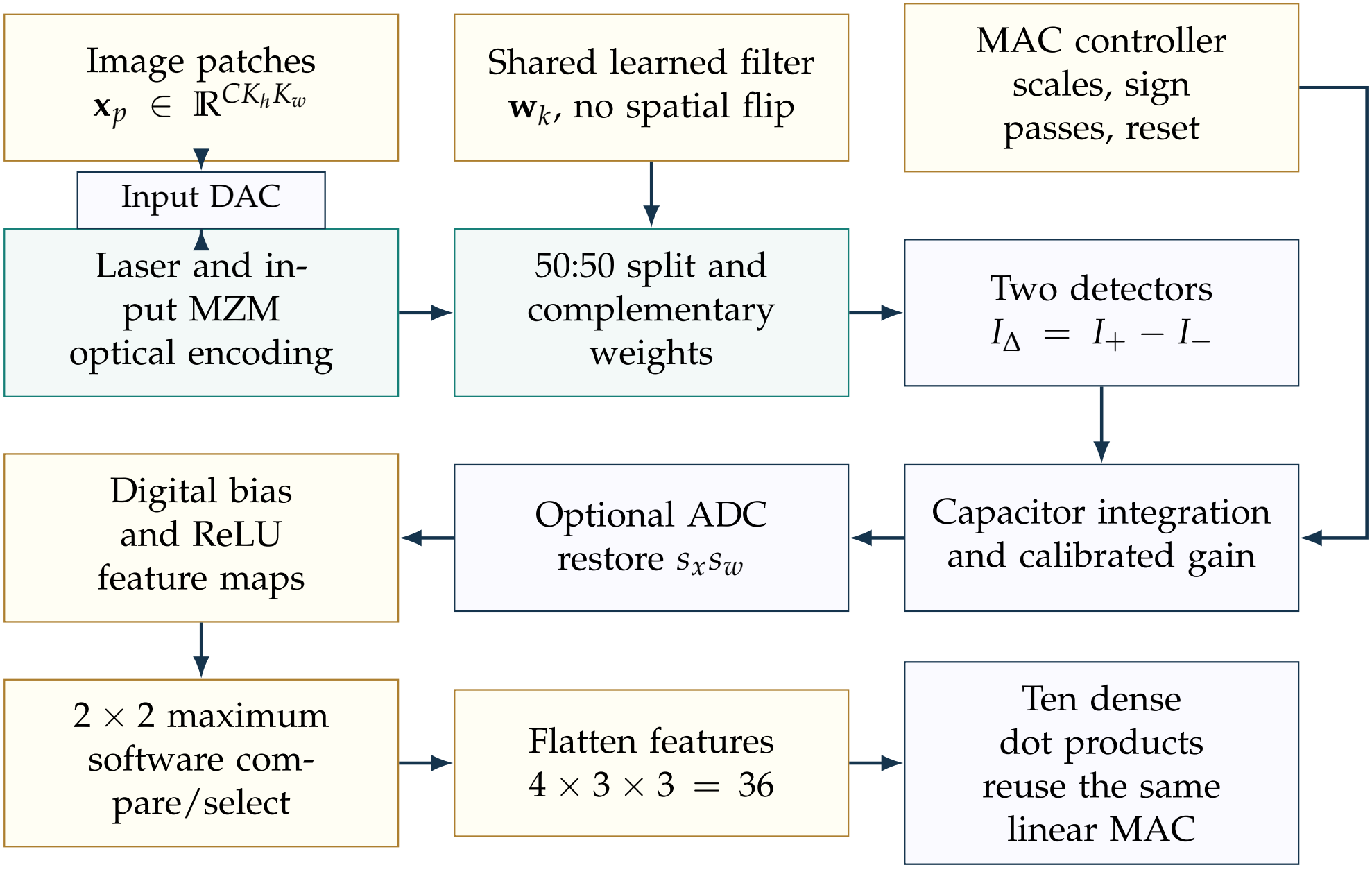



Teal: optical path. Blue: electrical accumulation/readout. Gold: software/control.
Convolutional layer: 36 patches $\times$ 4 filters = 144 nine-input MACs.
Dense layer: 10 MACs $\times$ 36 inputs. Final digital bias and argmax select the digit.
The complete ANN neuron adds sigmoid and ADC; CNN filters reuse its linear core.